

SREYA S

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Professional Summary

B.Tech. Computer Science and Engineering student with a strong interest in Machine Learning and Natural Language Processing, with hands-on experience building ML pipelines and conducting Named Entity Recognition (NER) research as a Research Intern at IIT Palakkad. Passionate about developing data-driven solutions to real-world problems.

Education

NSS College of Engineering

2023 – 2027

B.Tech in Computer Science & Engineering — CGPA: 9.32

Experience

Research Intern — Building Indic models for named entity Recognition

June 2026

IIT Palakkad — Research Internship, Natural Language Processing

- Completed a research internship in the Data Science department at IIT Palakkad, working on a Natural Language Processing project focused on low-resource Indian languages.
- Applied machine learning and NLP techniques to build and evaluate data pipelines, gaining hands-on experience with model training, evaluation, and research documentation.

Projects

FileSense – Context-Aware Intelligent File Organizer | [github](#)

Academic Project (6 months)

- Architected a context-aware intelligent file organization system in Python using NLP, machine learning, and Sentence-Transformer (BERT-based) model for semantic search and file classification.
- Implemented embedding-based contextual tagging and a semantic search engine using cosine similarity over sentence embeddings, enabling retrieval by meaning rather than exact filename matches.
- Built multi-format text extraction (PDF, DOCX, TXT) and a hybrid priority-ranking algorithm.

Audio Filtering Pipeline for TTS Training Data | [github](#)

- Built a distributed audio data quality filtering pipeline, processing the IndicVoices dataset to filter out low-quality samples before Text-to-Speech model training.
- Implemented multi-metric quality checks — clipping ratio, SNR, duration, and speech ratio — using librosa-based signal processing, alongside language verification via a SpeechBrain ECAPA-TDNN (VoxLingua107) model.
- Designed a fault-tolerant, parallelized architecture using Ray for distributed execution across CPU cores, with a resumable JSONL manifest tracking KEEP/DISCARD decisions, plus a Whisper-based ASR intelligibility (CER) scoring signal.

Skills

Languages: Python, C, Java, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Hugging Face Transformers, Sentence-Transformers

Databases & Storage: MySQL, SQLite

Tools: Git, GitHub, Linux (Ubuntu)

Concepts: Object Oriented Programing, NLP, NER, LLM, DSA, DBMS

Additional Information

Certifications: Agentic AI: From Learner to Builder – IBM SkillsBuild (CSRBOX) , NPTEL: The Joy of Computing using Python IIT Madras 93% (Elite)